Spatial Degradation of Hepatocyte Zonation
Ruler Battery — Per-Set Report

Auto-generated

2026-06-22

Method (read first)

Candidate zonation rulers were built, each scored as a per-cell coordinate (coord = mean_z(PC) − mean_z(PP), or a learned projection). **Ruler quality is judged ONLY on Paper 2 / healthy-atlas criteria** (8-marker validation gate, healthy PC–PP anticorrelation, split-half reproducibility). **Two co-primary rulers** are frozen — an interpretable published anchor and a label-free learned axis — then transferred to the Paper 1 disease cohort, where H1/H2/H3 are TEST results — never used for selection. Eligibility for the primary slot is *leakage-clean* (axis NOT fit on Paper-1 cells), *not* publishability: rulers fit on Paper-1 healthy cells (`unsupervised`, `unsupervised_combined`) have in-sample healthy metrics, so they are shown as *control (Paper1-fit)* and excluded from selection while still serving as H1 robustness checks. Sections below are ordered by healthy-ruler quality (split-half + $\max(0, -\text{anticorr})$). H1 = donor-level zonation collapse; H2 = zonal-slope loss; H3 = de-zonation/plasticity coupling.

Selection decision.

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CO-PRIMARY rulers (frozen on HEALTHY metrics only; eligibility = leakage-clean, i.e. NOT fit on Paper-1

  [interpretable / published anchor] expanded_curated: score=+1.149, anticorr=-0.479, split-half=+0.670
  [label-free / learned, external to Paper 1] unsupervised_p2: score=+1.162, anticorr=-0.471, split-hal

Highest healthy score among ALL eligible rulers: unsupervised_p2.
Interpretable vs label-free healthy-score gap = 0.012 (within the 0.02 tie band) -- reported side-by-si

Excluded from the competition (fit on Paper-1 cells -> in-sample healthy metrics, kept as H1 robustness
selected_frozen.txt = copy of the interpretable anchor (for downstream H2b/H2c).
WARNING: selection used HEALTHY-atlas criteria ONLY, NOT Paper 1 disease effect (disease is the transfe
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unsupervised_combined control (Paper1-fit)

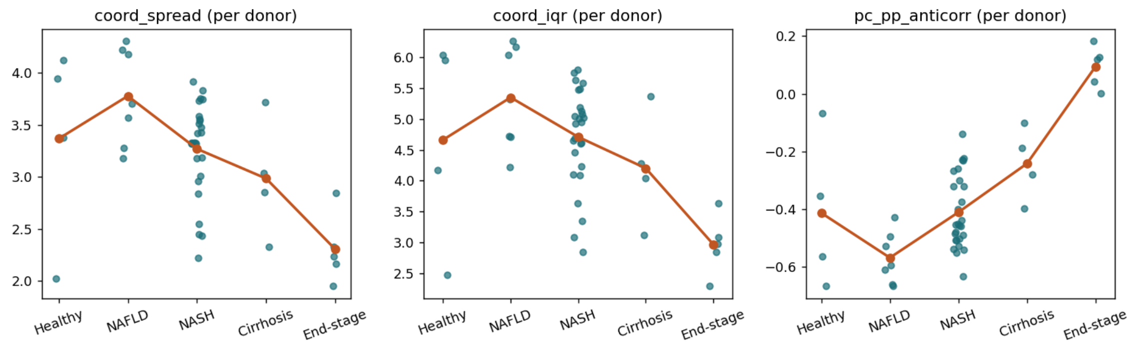
Candidate zonation ruler. **Genes:** 4769 PC + 3719 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.498 (want strongly negative); split-half reproducibility = +0.720; coherence PC/PP = +0.087/+0.057.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.514 (perm p = 0.0005); PC–PP anticorr trend rho = +0.625 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 100 genes tested; 86% weaken; median trend rho = -0.258.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = -0.010; 24/47 donors >0; sign-test p = 1.



unsupervised_p2 PRIMARY (label-free)

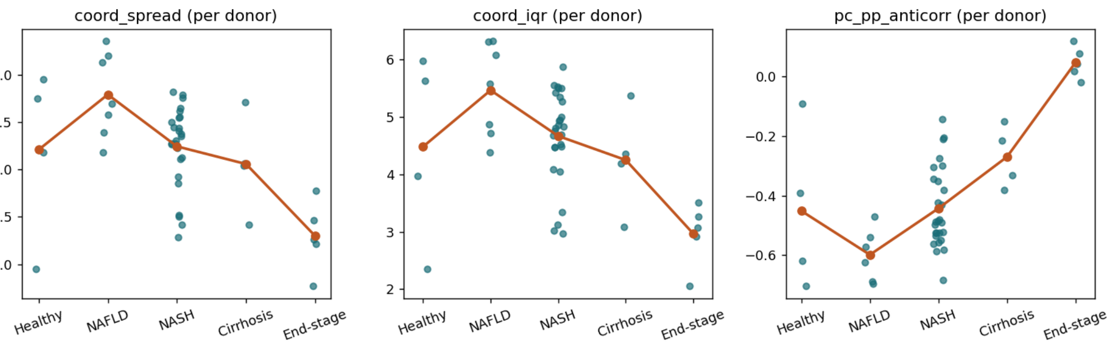
Candidate zonation ruler. **Genes:** 5177 PC + 3302 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.471 (want strongly negative); split-half reproducibility = +0.691; coherence PC/PP = +0.076/+0.052.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.504 (perm p = 0.0005); PC–PP anticorr trend rho = +0.635 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 100 genes tested; 83% weaken; median trend rho = -0.260.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = -0.009; 23/47 donors >0; sign-test p = 1.



expanded_curated PRIMARY (interpretable)

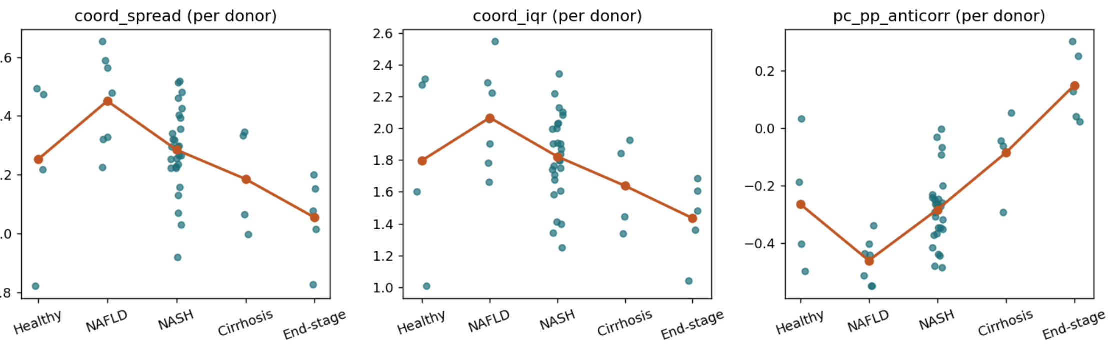
paper2_landmark plus curated extra anchors (26+23), de-duplicated. **Genes:** 26 PC + 23 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.479 (want strongly negative); split-half reproducibility = +0.670; coherence PC/PP = +0.155/+0.065.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.434 (perm p = 0.003); PC–PP anticorr trend rho = +0.612 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 49 genes tested; 90% weaken; median trend rho = -0.258.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.025; 33/47 donors >0; sign-test p = 0.0079.



selected_frozen primary_candidate

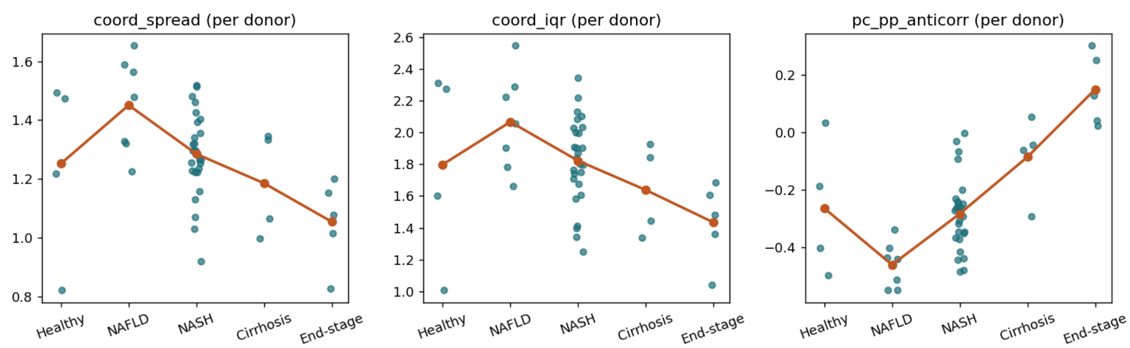
Copy of the interpretable co-primary anchor (best leakage-clean PUBLISHED set by healthy metrics); drives downstream H2b/H2c. **Genes:** 26 PC + 23 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.479 (want strongly negative); split-half reproducibility = +0.670; coherence PC/PP = +0.155/+0.065.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.434 (perm p = 0.003); PC–PP anticorr trend rho = +0.612 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 49 genes tested; 90% weaken; median trend rho = -0.258.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.025; 33/47 donors >0; sign-test p = 0.0079.



unsupervised_top100 primary_candidate

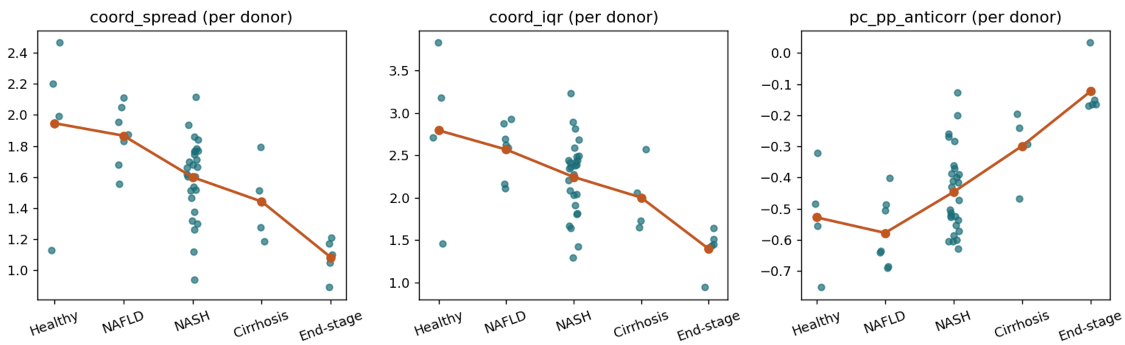
The 100+100 strongest +/- loading genes of the unsupervised PCA axis. **Genes:** 100 PC + 100 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.292 (want strongly negative); split-half reproducibility = +0.821; coherence PC/PP = +0.088/+0.056.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.608 (perm p = 0.0005); PC–PP anticorr trend rho = +0.613 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 200 genes tested; 92% weaken; median trend rho = -0.324.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.031; 36/47 donors >0; sign-test p = 0.00035.



unsupervised_top50 primary_candidate

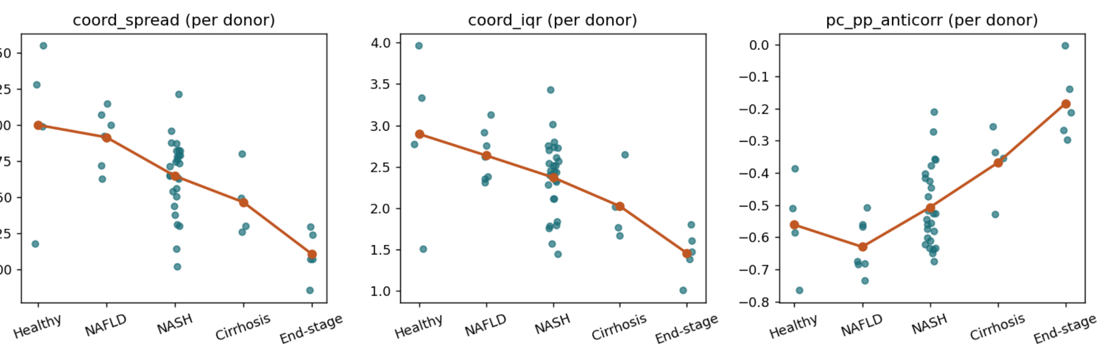
The 50+50 strongest +/- loading genes of the unsupervised PCA axis, scored as a signature. **Genes:** 50 PC + 50 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.306 (want strongly negative); split-half reproducibility = +0.785; coherence PC/PP = +0.121/+0.082.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.617 (perm p = 0.0005); PC–PP anticorr trend rho = +0.607 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 100 genes tested; 96% weaken; median trend rho = -0.372.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.027; 38/47 donors >0; sign-test p = 2.5e-05.



unsupervised control (Paper1-fit)

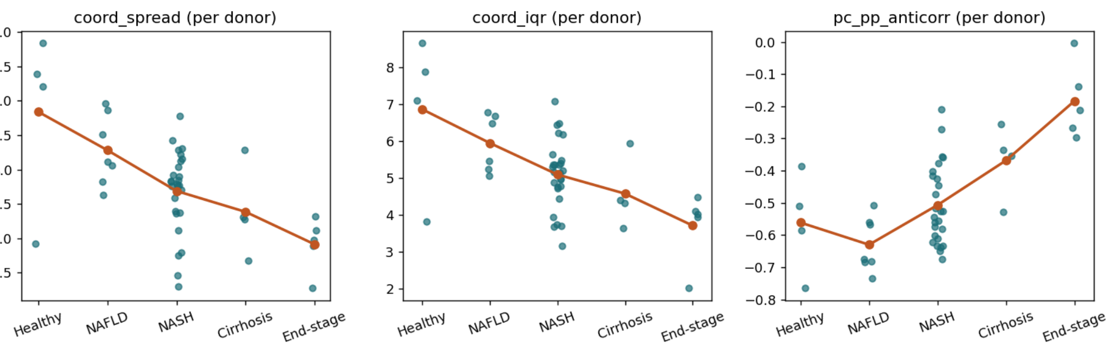
Label-free PCA porto-central axis learned on Paper 1 HEALTHY cells (sensitivity version). **Genes:** 4585 PC + 3903 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 7/8 (PASS). Healthy PC–PP anticorrelation = -0.306 (want strongly negative); split-half reproducibility = +0.781; coherence PC/PP = +0.121/+0.082.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.562 (perm p = 0.0005); PC–PP anticorr trend rho = +0.607 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 100 genes tested; 96% weaken; median trend rho = -0.375.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.016; 32/47 donors >0; sign-test p = 0.019.



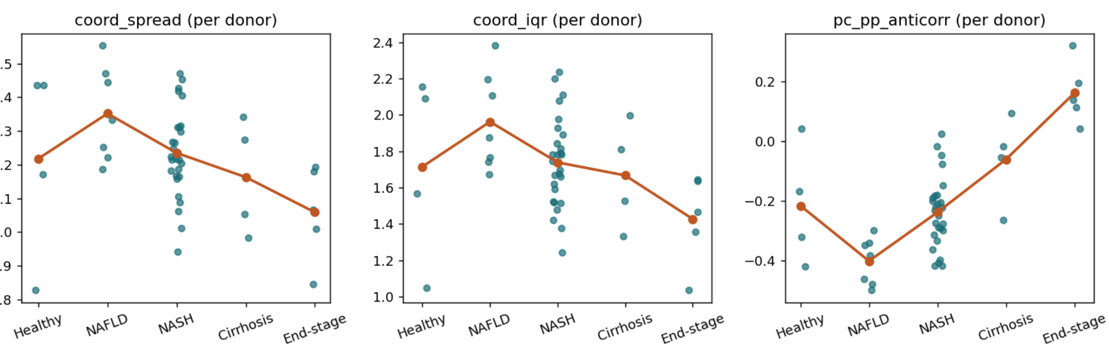
Paper 2's exact 20+20 hepatocyte landmark genes (Hepatocyte-{PC,PP}-LM.csv), the genes Paper 2 itself uses to assign per-cell snRNA zonation (eta method). **Genes:** 20 PC + 20 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.455 (want strongly negative); split-half reproducibility = +0.631; coherence PC/PP = +0.156/+0.062.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.380 (perm p = 0.0085); PC–PP anticorr trend rho = +0.608 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 40 genes tested; 78% weaken; median trend rho = -0.229.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.024; 33/47 donors >0; sign-test p = 0.0079.



unsupervised_top250 primary_candidate

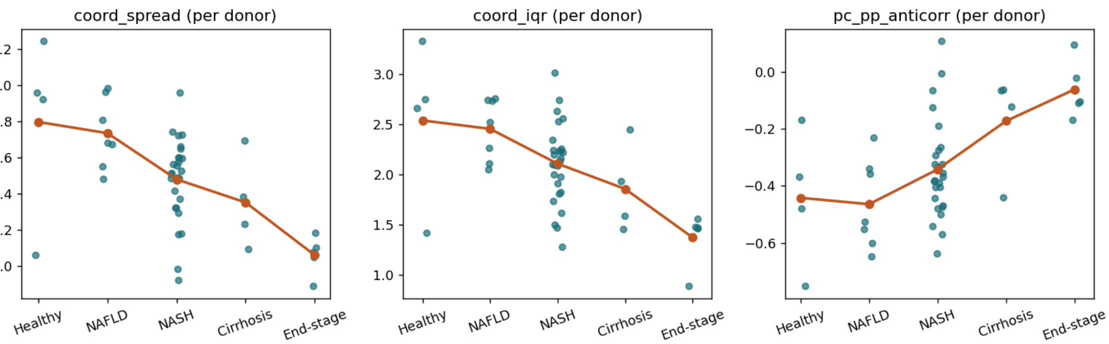
Candidate zonation ruler. **Genes:** 250 PC + 250 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.227 (want strongly negative); split-half reproducibility = +0.853; coherence PC/PP = +0.047/+0.033.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.575 (perm p = 0.0005); PC–PP anticorr trend rho = +0.526 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 500 genes tested; 81% weaken; median trend rho = -0.241.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.025; 31/47 donors >0; sign-test p = 0.04.



paper2_top50 primary_candidate

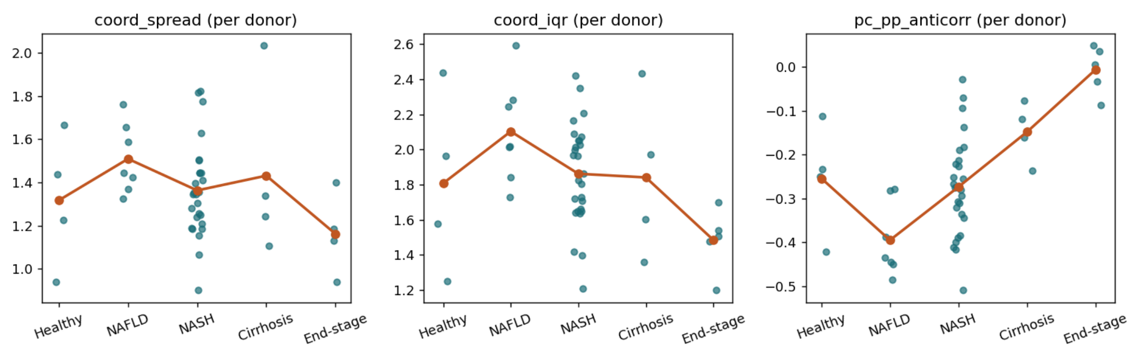
Top 50+50 genes by —center-of-mass— among q¡0.05 zonated genes (supplementary table 8). **Genes:** 50 PC + 50 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.404 (want strongly negative); split-half reproducibility = +0.573; coherence PC/PP = +0.051/+0.034.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.316 (perm p = 0.035); PC–PP anticorr trend rho = +0.564 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 88 genes tested; 73% weaken; median trend rho = -0.204.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.010; 33/47 donors >0; sign-test p = 0.0079.



paper2_top100 primary_candidate

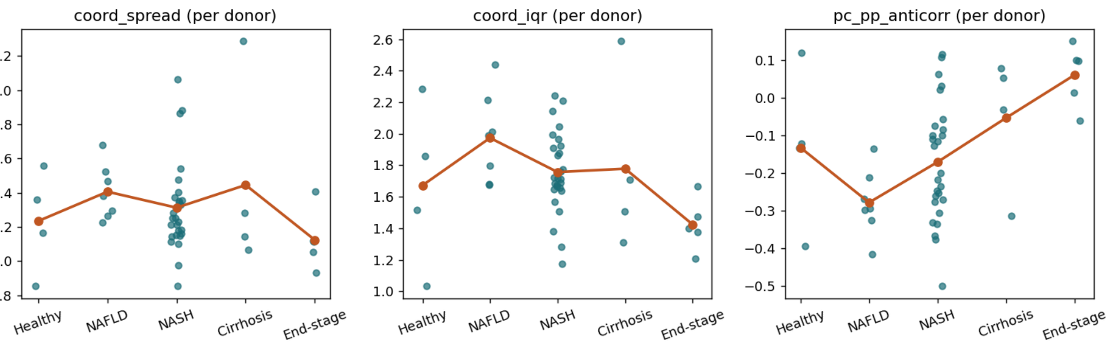
Top 100+100 zonated genes by —center-of-mass—. **Genes:** 100 PC + 100 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.273 (want strongly negative); split-half reproducibility = +0.563; coherence PC/PP = +0.028/+0.020.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.284 (perm p = 0.06); PC–PP anticorr trend rho = +0.423 (perm p = 0.003).

H2 (zonal slope-loss, held-out). 186 genes tested; 59% weaken; median trend rho = -0.066.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.011; 28/47 donors >0; sign-test p = 0.24.



unsupervised_full exploratory

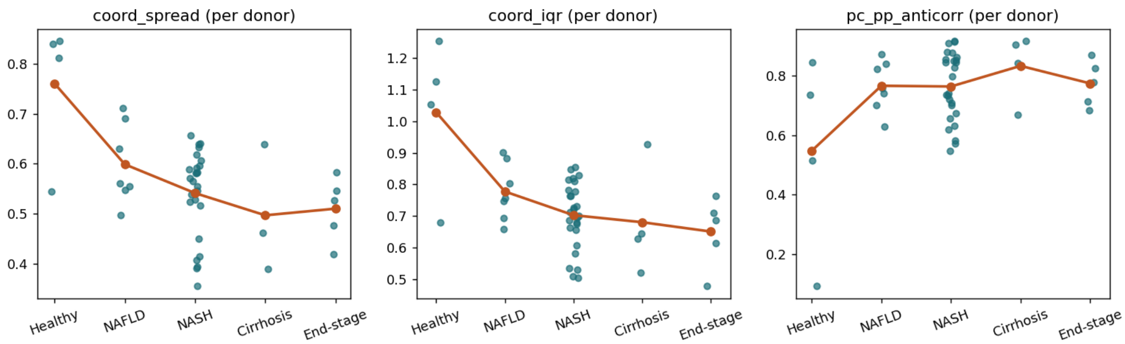
Candidate zonation ruler. **Genes:** 4585 PC + 3903 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 6/8 (PASS). Healthy PC–PP anticorrelation = +0.646 (want strongly negative); split-half reproducibility = +0.831; coherence PC/PP = +0.008/+0.011.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.421 (perm p = 0.0035); PC–PP anticorr trend rho = +0.196 (perm p = 0.2).

H2 (zonal slope-loss, held-out). 8488 genes tested; 61% weaken; median trend rho = -0.048.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = -0.008; 21/47 donors >0; sign-test p = 0.56.



core_curated **interpretability_only**

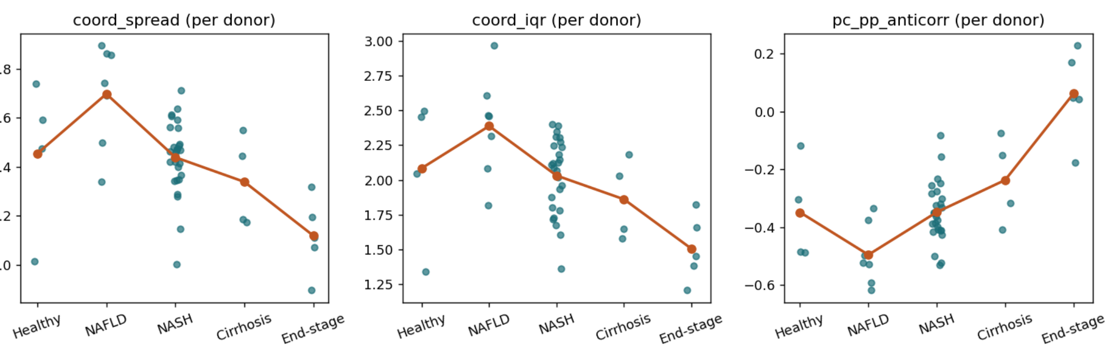
13+8 hand-curated, highly interpretable canonical anchors (GLUL, CYP2E1, ASS1, HAL, ...). **Genes:** 13 PC + 8 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.278 (want strongly negative); split-half reproducibility = +0.537; coherence PC/PP = +0.185/+0.104.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.541 (perm p = 0.0005); PC–PP anticorr trend rho = +0.562 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 21 genes tested; 100% weaken; median trend rho = -0.339.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.033; 36/47 donors >0; sign-test p = 0.00035.



unsupervised_lasso robustness

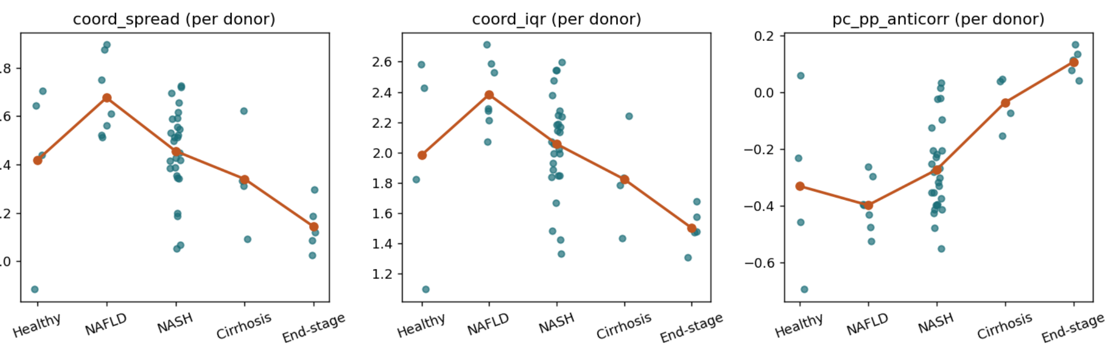
Lasso (L1) sparse ruler distilling the Paper-2-trained PCA axis (clean/external training). **Genes:** 126 PC + 149 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.184 (want strongly negative); split-half reproducibility = +0.629; coherence PC/PP = +0.022/+0.020.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.505 (perm p = 0.0005); PC–PP anticorr trend rho = +0.602 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 275 genes tested; 69% weaken; median trend rho = -0.144.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = -0.017; 22/47 donors >0; sign-test p = 0.77.



unsupervised_elasticnet robustness

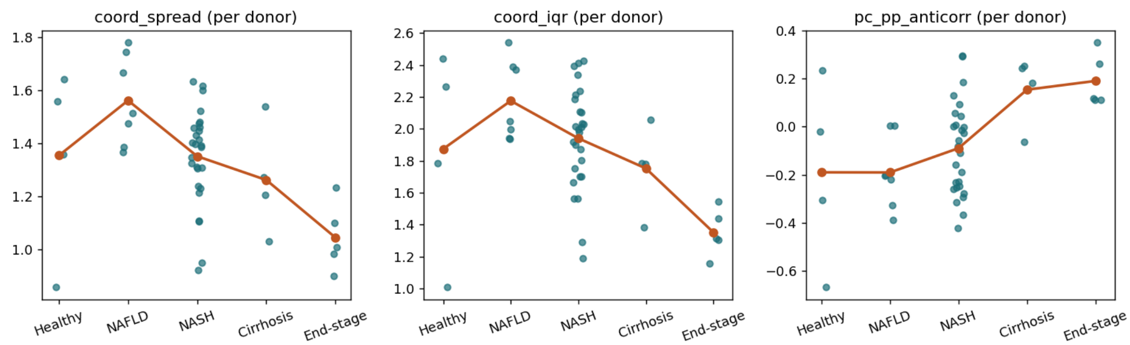
Elastic-net (L1+L2) sparse ruler distilling the Paper-2-trained PCA axis (clean). **Genes:** 325 PC + 281 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.068 (want strongly negative); split-half reproducibility = +0.689; coherence PC/PP = +0.016/+0.014.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.492 (perm p = 0.0005); PC–PP anticorr trend rho = +0.483 (perm p = 0.0025).

H2 (zonal slope-loss, held-out). 606 genes tested; 67% weaken; median trend rho = -0.121.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = -0.020; 21/47 donors >0; sign-test p = 0.56.



paper2_top250 exploratory

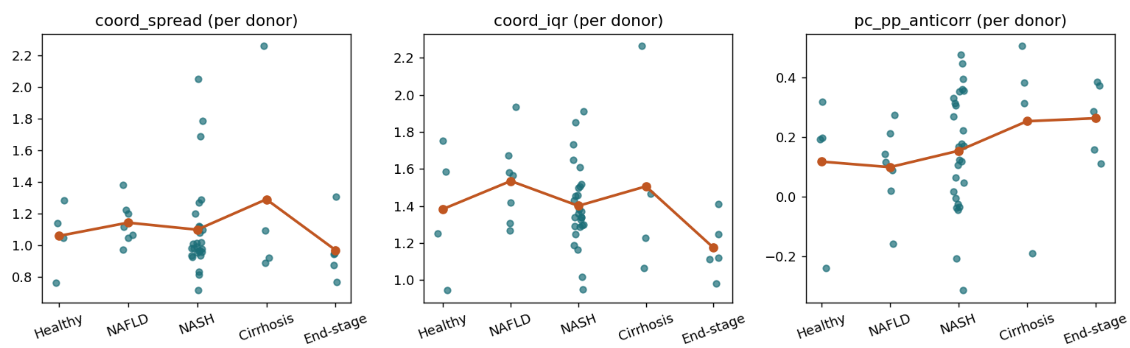
Top 250+250 zonated genes by —center-of-mass—. **Genes:** 250 PC + 250 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = +0.041 (want strongly negative); split-half reproducibility = +0.554; coherence PC/PP = +0.015/+0.011.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.265 (perm p = 0.067); PC–PP anticorr trend rho = +0.250 (perm p = 0.081).

H2 (zonal slope-loss, held-out). 483 genes tested; 53% weaken; median trend rho = -0.011.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.006; 26/47 donors >0; sign-test p = 0.56.



paper2_full exploratory

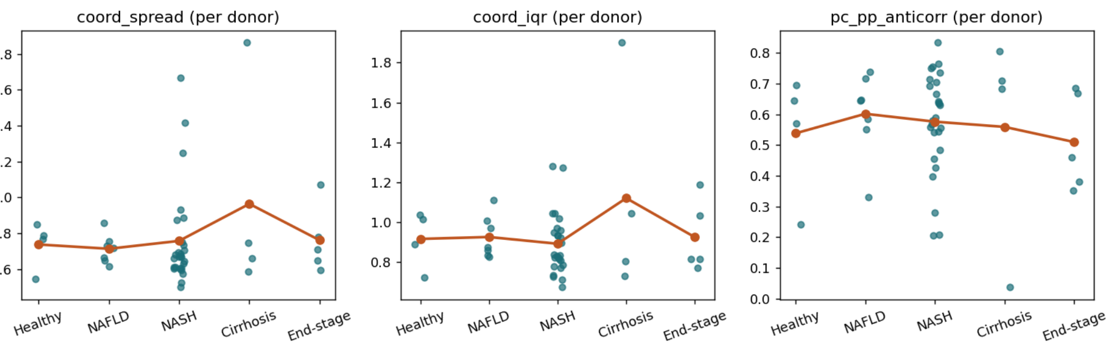
All significant zonated hepatocyte genes (q¡0.05, max-expr¿1e-4), split by COM sign (1624+447). **Genes:** 1624 PC + 447 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = +0.556 (want strongly negative); split-half reproducibility = +0.484; coherence PC/PP = +0.005/+0.009.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = -0.042 (perm p = 0.77); PC–PP anticorr trend rho = -0.040 (perm p = 0.79).

H2 (zonal slope-loss, held-out). 2035 genes tested; 41% weaken; median trend rho = +0.037.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.000; 25/47 donors >0; sign-test p = 0.77.



full exploratory

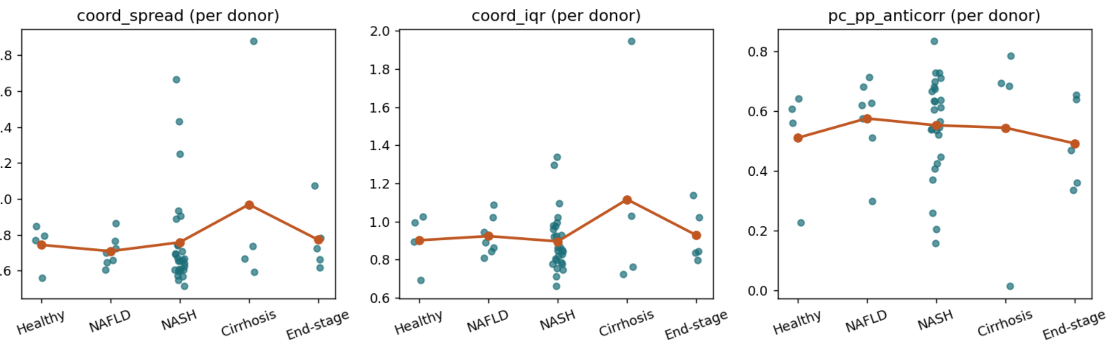
Legacy transcriptome-wide unweighted signature mean (main pipeline). **Genes:** 0 PC + 0 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = +0.532 (want strongly negative); split-half reproducibility = +0.464; coherence PC/PP = +0.008/+0.010.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = +0.021 (perm p = 0.88); PC–PP anticorr trend rho = +0.007 (perm p = 0.97).

H2 (zonal slope-loss, held-out). 1637 genes tested; 42% weaken; median trend rho = +0.034.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.002; 26/47 donors >0; sign-test p = 0.56.



supervised exploratory

Whole-transcriptome multinomial logistic regression trained on Paper 2 zone labels (landmarks excluded). **Genes:** 4504 PC + 3935 PP.

Validation (8-marker healthy gate) & controls. Markers correct: 8/8 (PASS). Healthy PC–PP anticorrelation = -0.005 (want strongly negative); split-half reproducibility = +0.452; coherence PC/PP = +0.064/+0.014.

H1 (donor-level collapse, TRANSFER). coord-spread vs stage rho = +0.122 (perm p = 0.4); PC–PP anticorr trend rho = +0.569 (perm p = 0.0005).

H2 (zonal slope-loss, held-out). 100 genes tested; 87% weaken; median trend rho = -0.169.

H3 (de-zonation ↔ plasticity, donor-level). mean rho = +0.003; 27/47 donors >0; sign-test p = 0.38.